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# 01_setup_download.py
# Google Colab: Runtime > Change runtime type > T4 GPU (GPU is needed later, not here).
# Run each "# %%" block as its own Colab cell, top to bottom.
```

```
# %% [1] Install packages (once per Colab session)
import os, glob, json, subprocess, zipfile, tarfile


def sh(cmd):
    print("$", cmd)
    subprocess.run(cmd, shell=True, check=True)


sh("pip -q install kaggle py7zr tifffile")
```

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# %% [2] Mount Google Drive (results live on Drive so they survive session resets)
ROOT = os.environ.get("AMAZON_ROOT", "/content/drive/MyDrive/amazon_project")
RAW = os.environ.get("AMAZON_RAW", "/content/data")
try:
    from google.colab import drive
    drive.mount("/content/drive")
except ImportError:
    print("Not running on Colab: using local folders instead")
os.makedirs(ROOT, exist_ok=True)
os.makedirs(RAW, exist_ok=True)
```

```
# %% [3] Kaggle credentials
# Kaggle > profile > Settings > API > "Create New Token" gives you kaggle.json.
# If Kaggle shows a token string instead, follow the instructions printed on that page.
if not os.path.exists(os.path.expanduser("~/.kaggle/kaggle.json")):
    from google.colab import files
    files.upload()  # choose kaggle.json
    sh("mkdir -p ~/.kaggle && mv kaggle.json ~/.kaggle/ && chmod 600 ~/.kaggle/kaggle.json")
```

```
# %% [4] Accept the competition rules (one-time, in your browser)
# Open https://www.kaggle.com/competitions/planet-understanding-the-amazon-from-space
# Click "Late Submission" / "I Understand and Accept". Without this, downloads return 403.
COMP = "planet-understanding-the-amazon-from-space"
sh(f"kaggle competitions files -c {COMP}")
# Look at the printed file names. You need the labels CSV and the 4-band TIF training images.
```

```
# %% [5] Download (edit FILES if the names printed above differ)
# The TIF archive is large (many GB). Make sure Colab shows plenty of free disk space.
FILES = ["train_v2.csv", "train-tif-v2.tar.7z"]
for f in FILES:
    sh(f"kaggle competitions download -c {COMP} -f {f} -p {RAW}")
```

```
# %% [6] Extract
import py7zr

for z in glob.glob(f"{RAW}/*.zip"):
    with zipfile.ZipFile(z) as zf:
        zf.extractall(RAW)
for a in glob.glob(f"{RAW}/*.7z"):
    print("Extracting", a, "(this can take 20+ minutes)")
    with py7zr.SevenZipFile(a) as sz:
        sz.extractall(RAW)
for t in glob.glob(f"{RAW}/**/*.tar", recursive=True):
    print("Untarring", t)
    with tarfile.open(t) as tf:
        tf.extractall(RAW)
```

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# %% [7] Verify and record paths for the next scripts
tifs = glob.glob(f"{RAW}/**/*.tif", recursive=True)
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csvs = glob.glob(f"{RAW}/**/train_v2.csv", recursive=True)
print("TIF files found:", len(tifs), "(expected 40479)")
assert len(tifs) > 1000 and csvs, "Extraction failed: check the file names in step 4"
paths = {"tif_dir": os.path.dirname(tifs[0]), "csv": csvs[0]}
with open(f"{ROOT}/paths.json", "w") as f:
    json.dump(paths, f)
print(paths)
print("Done. Next: run 02_eda_and_cache.py")
```